

On clausal complementation, once more

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Abstract

This paper develops a novel analysis of clausal complementation based on the distribution of Greek declarative clauses introduced by *oti* and *pu*. These clauses present three puzzles: (i) they are in complementary distribution after most verbs; (ii) they are licensed as internal arguments and derived subjects but are banned from the external argument position; and (iii) complement *pu*-clauses impose a previously unnoticed stativity restriction on the matrix verb, a restriction absent in their adjunct uses. I argue that (ii) and (iii) receive a unified explanation if *oti* and *pu* of complement clauses select a light noun in their specifier (see also Arsenijević 2009; Moltmann 2019, 2024). This noun must be licensed via incorporation into the matrix verb (Hale and Keyser 1993), an operation that succeeds from the complement position but fails from the external argument position, thus explaining the internal-external argument asymmetry. The semantic distinction between content (*oti*) and situation (*pu*) explains their distribution after verbs (see also Bondarenko 2022; Elliott 2020; Kratzer 2006; Moltmann 1989, 2013a,b, 2021; Moulton 2009, 2015). Finally, the stativity restriction reveals a distinction in the way arguments and adjuncts are licensed (Bruening 2013; Hunter 2015; Zeijlstra 2020; Hewett 2023; Neeleman et al. 2023, i.a.). Beyond resolving the puzzles in Greek, this analysis settles the open question of whether embedded clauses occupy the same argument position as DPs. Drawing on multiple diagnostics, I show that they can, thereby directly challenging approaches (e.g., Bondarenko 2022) that deny this possibility. The analysis carries broader implications for the role of selection in clausal complementation, the syntax of nominalized clauses, and the Syntax-Semantics interface.

1 Introduction

The syntax of clausal complementation has long been governed by the view that embedded clauses are licensed by the selectional properties of the matrix predicate—that is, the predicate selects either the category or the semantic content of the clause it embeds (Chomsky 1965; Bresnan 1972; Grimshaw 1979, though see Adger and Quer 2001 i.a.). This paper challenges this foundational assumption through an investigation of declarative embedded clauses in Modern Greek introduced by *oti* and *pu*. These clauses present three puzzles for theories of clausal complementation.

First, *oti* and *pu* are in complementary distribution after most verbs. Verbs of saying and belief

are compatible with *oti*, while emotive factive verbs are compatible *pu*, as illustrated in (1).^{1,2}

- (1) a. I Elena ipe *oti*/**pu* exi episkefti tin
the.F.SG.NOM Elena.F.SG.NOM said.3SG OTI/PU have.3SG visited the.F.SG.ACC
Vrazilia.
Brazil.F.SG.ACC
‘Elena said that she has visited Brazil.’
b. I Elena metanjose *pu*/**oti* paretiθike.
the.F.SG.NOM Elena.F.SG.NOM regretted.3SG PU/OTI quit.3SG
‘Elena regretted that she quit.’

Their distribution cannot be simply reduced to selectional properties of predicates, whether semantic (e.g., familiarity, factivity), or categorial (e.g., CP, DP, PP) (Christidis 1982; Angelopoulos 2019).

Second, *oti*- and *pu*-clauses exhibit an asymmetry in their distribution as arguments. While they function freely as internal arguments and can surface as derived subjects (e.g., in passives), they are banned from the external argument position.

Third, this paper brings to light a novel empirical fact: complement *pu*-clauses impose a strict stativity requirement on the matrix predicate. This restriction, however, vanishes when *pu* introduces adjunct clauses or is used as the interrogative pronoun *where*.

To solve these puzzles, I propose a rethinking of the selectional relationship between verb and clause. I argue that the selection is reversed: *oti* and *pu* are probes that select a light noun (Hale and Keyser 1993).³ This noun must then undergo incorporation into a higher verbal head thus forming the matrix verb, as proposed in Arsenijević (2009). This analysis derives the argument asymmetry directly: incorporation of the light noun from the complement position is licit, but incorporation from the external argument position is not possible. A crucial aspect of the analysis is that *oti*- and *pu*-clauses are merged in the verb’s complement position, occupying the same position as the corresponding DP arguments. This claim is supported by multiple well-established diagnostics. Consequently, analyses which deny that embedded clauses can occupy the syntactic position of DP arguments cannot be maintained.

The complementary distribution of *oti* and *pu* is derived from a semantic distinction between the types of clauses they introduce—*content* versus *situation*, respectively—which must match

¹ Glossing abbreviations: 1=first person, 2=second person, 3=third person, SG=singular, PL=plural, F=feminine, M=male, NOM=nominative, ACC=accusative, DAT=dative. In the examples throughout the paper, OTI and PU are used as placeholder glosses for the corresponding Greek complementizers.

² The judgments presented in this paper were verified with twelve native speakers of Modern Greek. We did not detect any cases of speaker variation in the judgments presented here.

³ I remain neutral about the categorial status of *oti* and *pu* throughout this paper. Previous studies that have explored the categorial status of these elements analyze them either as Cs (Roussou 1994) or Ds (Roussou 2010, 2019 i.a.). However, the debate over the categorial status of *oti* and *pu* appears to be of little consequence in these works, as the syntactic structure proposed for *oti*- and *pu*-clauses remains unchanged regardless of whether they are treated as Cs or Ds. Likewise, the analysis I propose here is not dependent on whether *oti* and *pu* are analyzed as Ds or something else, so I will not delve further into that debate (see also Baunaz and Lander 2018).

the semantics of the incorporating noun. Moreover, the stativity restriction is argued to follow from differences in how selectional dependencies are established in adjunct uses of *pu* (e.g., as the interrogative pronoun *where*) versus its use in argument clauses.

This paper is structured as follows. Section 2 lays out the empirical landscape, detailing the argument status of *oti/pu*-clauses, the internal-external argument asymmetry, and the novel stativity restriction. Section 3 develops the core light-noun incorporation analysis. Section 4 extends this analysis to derive the argument/adjunct distinction and solve the stativity puzzle. Section 5 provides corroborating evidence from nominalized clauses. Section 6 argues against alternative accounts, and Section 7 explores the broader theoretical implications of the proposed analysis. Section 8 concludes.

2 The empirical landscape of *oti*- and *pu*-clauses

This section establishes three core empirical findings about *oti*- and *pu*-clauses: (i) they function as genuine internal arguments in syntax, rather than adjuncts—as has been argued for embedded clauses in other languages (Bochnak and Hanink 2022)—in that they are selected by the predicate and occupy its theta-position, (ii) they are banned from the external argument position, and (iii) *pu*-complements are subject to a previously unnoticed stativity restriction.

2.1 *Oti*- and *pu*-clauses can be internal arguments

Three standard diagnostics—clitic doubling, passivization, and extraction—confirm that *oti*- and *pu*-clauses function as internal arguments.

First, clitic doubling in Greek is a reliable diagnostic for argumenthood, as it is restricted to arguments and unavailable for adjuncts. The examples in (2) illustrate this distinction. In (2a), the verb *majirepsa* ‘I cooked’ appears with an optional DP object, *tin pitsa* ‘the pizza,’ and a temporal DP adjunct, *tin nixta* ‘the night.’ However, when a clitic is introduced in (2b), it can only refer to the argument, *tin pitsa*, as just as in clitic doubling of the DP in (2c). The clitic cannot double the adjunct, which is why the intransitive interpretation ‘I cooked at night’ is unavailable.⁴

- (2) a. Majirepsa (tin pitsa) tin nixta.
 cooked.1SG the.F.SG.ACC pizza.F.SG.ACC the.F.SG.ACC night.F.SG.ACC
 ‘I cooked the pizza at night.’
- b. Tin_{i/j} majirepsa tin nixta_i.
 3SG.F.ACC cooked.1SG the.F.SG.ACC night.F.SG.ACC
 ‘I cooked it at night.’
- c. Tin_{i/j} majirepsa tin pitsa_i tin nixta_j.
 3SG.F.ACC cooked.1SG the.F.SG.ACC pizza.F.SG.ACC the.F.SG.ACC night.F.SG.ACC
 ‘I cooked the pizza at night.’

⁴ In Angelopoulos and Sportiche (2021), clitic doubling is an Agree relation, and clitics are agreement markers. In Greek, there is no agreement with adjuncts, which explains why arguments only can be clitic-doubled.

Applying this diagnostic to clausal complements reveals their argumental status. As shown in (3), both *oti*- and *pu*-clauses pattern with arguments in that they can be doubled by the neuter clitic *to*.

- (3) a. To_i $pistevi$ [$oti/*pu$ $katalaveni$ tin $katastasi$] $_i$.
 3SG.N.ACC believe.3SG OTI/*PU understand.3SG the.F.SG.ACC situation.F.SG.ACC
 ‘She believes that he understands the situation.’
 b. To_i $metanjose$ [$pu/*oti$ $\delta en \theta a$ $ksanavlepodan pote$] $_i$.
 3SG.N.ACC regret.3SG PU/*OTI not would see.3PL never
 ‘She regretted that they would never see each other again.’

A comparison with nominalized clauses is needed to rule out the presence of a silent D on top of clitic-doubled *oti*-clauses.⁵ This cannot be shown with *pu*-clauses because, as we will see in Section 5, they cannot be nominalized. The two constructions are semantically distinct: in active sentences such as (4a), the bare *oti*-clause provides the explanation for the event or state of affairs expressed by the matrix verb (*explanans*). On the other hand, the nominalized *oti*-clause express the fact being explained (*explanandum*) (Kastner 2015; Elliott 2020 i.a.). (4c) and (4d) show that this semantic difference is maintained when a bare or a nominalized clauses are clitic-doubled. If clitic-doubled *oti*-clauses were embedded under a silent D, they would be DPs and would be expected to pattern with the overt nominalized clause in (4c), yielding an *explanandum* reading. Since they do not, this shows that bare *oti*-clauses are true internal arguments, and not DPs in disguise.

- (4) a. I Maria mas eksijise kala [oti i
 the.F.SG.NOM Maria.F.SG.NOM us.1PL.DAT explained.3SG well OTI the.F.SG.NOM
 Ji ine strogili].
 Earth.F.SG.NOM be.3SG round.F.SG.NOM
 ‘Maria explained well that the Earth is round.’ (*explanans*)
 b. I Maria mas eksijise kala [to oti
 the.F.SG.NOM Maria.F.SG.NOM us.1PL.DAT explained.3SG well the.N.SG.ACC OTI
 i Ji ine strogili].
 the.F.SG.NOM Earth.F.SG.NOM be.3SG round.F.SG.NOM
 ‘Maria explained the fact that the Earth is round.’ (*explanandum*)

⁵ Similar constructions—where a clause is doubled by a pronoun—also occur in Azeri. Bondarenko (2023) analyzes the pronoun *there* as an E-type trace of the nominalized clause, with the bare CP functioning as a predicate that restricts the quantifier. This analysis, however, does not extend to Greek. Greek accusative clitics, such as the one in (3), are not pronouns; rather, they are considered pure agreement markers without semantic content, base-generated in their surface position rather than derived through movement (Angelopoulos and Sportiche 2021, Paparounas and Salzmann 2024). Finally, if the pronoun and the clause formed a constituent, we would expect clitic-doubled clauses to have the *explanandum* interpretation because they would be turned into DPs—but this is not what we find.

- c. I Maria mas to_i eksijise kala [oti
the.F.SG.NOM Maria.F.SG.NOM us.1PL.DAT 3SG.N.ACC explained.3SG well OTI
i Ji ine strogili]_i.
the.F.SG.NOM Earth.F.SG.NOM be.3SG round.F.SG.NOM
'Maria explained well that the Earth is round.' (*explanans*)
- d. I Maria mas to_i eksijise kala
the.F.SG.NOM Maria.F.SG.NOM us.1PL.DAT 3SG.N.ACC explained.3SG well
[to oti i Ji ine strogili]_i.
the.N.SG.ACC OTI the.F.SG.NOM Earth.F.SG.NOM be.3SG round.F.SG.NOM
'Maria explained the fact that the Earth is round.' (*explanandum*)

Second, passivization provides further evidence. As shown in (6a), *oti*-clauses can serve as subjects of passivized verbs. If they originate as internal arguments, their movement to the subject position should be blocked by a higher dative unless that dative is clitic-doubled, a restriction independently observed with DPs (Anagnostopoulou 2003). Thus, in ditransitives like *eksigo* 'explain,' the dative intervenes unless clitic-doubled, in which case the theme DP can raise, (5). *Oti*-clauses pattern exactly like DP themes: (a) they passivize, unlike adjuncts, and (b) they show the same intervention effects. For instance, in (6a), an *oti*-clause raises to subject in the passive *eksijithike* 'was explained' only if the dative *tis donatas* 'to Donata' is doubled by *tis*. This confirms that *oti*-clauses undergo genuine passivization, in contrast to adjuncts, which cannot.⁶ Finally, as (6) shows, the *explanans* vs. *explanandum* distinction displayed by bare *oti*- and nominalized *oti*-clauses is preserved under passivization, suggesting no hidden D-layer on top of passivized *oti*-clauses (see also Morgunova and Tsilia 2024).⁷

- (5) a. Eksijisa tis donatas afto to
explained.1SG the.F.SG.DAT donata.F.SG.DAT this.N.SG.ACC the.N.SG.ACC
jeynos.
fact.N.SG.ACC
'I explained this fact to Donata.'

⁶ A reviewer asks why adjuncts cannot undergo passivization. Addressing this question in full would require adopting a specific theory of passivization, which lies beyond the scope of the present paper. For the sake of argument, however, let us assume the smuggling analysis of the passive proposed by Collins (2005). Under this approach, XPs contained within the VP can be smuggled past the external argument and promoted to Spec,TP. Since the internal argument is inside the VP, it can undergo this process; adjuncts, by contrast, are not contained within the VP, which explains why they resist passivization.

⁷ A reviewer raises the possibility of analyzing these structures with a silent expletive *pro*, analogous to the English sentence, *It was explained to us that Mary is sick*. While this is a plausible initial hypothesis, it faces two significant obstacles. First, there is a broad consensus in the literature that Greek lacks both overt and silent expletives (e.g., Alexiadou and Anagnostopoulou 1998 i.a.). Second, and more crucially, the syntactic behavior of the construction is incompatible with an expletive analysis. Theoretically, a semantically vacuous expletive cannot be assigned a theta-role and must therefore be externally merged in Spec,TP solely to satisfy the EPP. However, the data show that the subject position triggers intervention effects, requiring cliticization of the dative argument. This behavior is characteristic of an element that has been moved from an internal argument position. An expletive base-generated in Spec,TP should not trigger such effects. Therefore, an analysis based on an expletive *pro* is empirically untenable for these structures.

- b. Afto to jeyonos *(tis) eksijithike
 this.N.SG.NOM the.N.SG.NOM fact.N.SG.NOM 3SG.F.DAT was.explained.3SG
 tis donatas.
 the.F.SG.GEN donata.F.SG.DAT
 ‘This fact was explained to Donata.’
- (6) a. [Oti i Ji ine strogili] *(tis)
 OTI the.F.SG.NOM Earth.F.SG.NOM be.3SG round.F.SG.NOM 3SG.F.DAT
 eksijithike tis donatas.
 was.explained.3SG the.F.SG.DAT donata.F.SG.DAT
 ‘That the Earth is round was explained to Donata.’ (*explanans*)
- b. [To oti i Ji ine strogili]
 the.N.SG.NOM OTI the.F.SG.NOM Earth.F.SG.NOM be.3SG round.F.SG.NOM
 *(tis) eksijithike tis donatas.
 3SG.F.DAT was.explained.3SG the.F.SG.DAT donata.F.SG.DAT
 ‘The fact the Earth is round was explained to Donata.’ (*explanandum*)

Third, *oti*- and *pu*-clauses are transparent for A-bar extraction of arguments, unlike adjunct clauses in Greek, which block extraction. As shown in (7a) and (8a), object extraction from both clause types is fully grammatical. *Oti*-clauses also permit adjunct extraction, (7b), whereas *pu*-clauses act as weak islands, blocking adjunct extraction, (8b) (Varlokosta 1994 i.a.). Adjunct clauses behave differently, as they block both argument and adjunct extraction. This is illustrated in (9) with a temporal *otan* ‘when’ clause. All other adjunct clauses (e.g., reason clauses) behave similarly, uniformly blocking argument and adjunct extraction (Kotzoglou 2025). In this respect, *oti/pu*-clauses pattern more closely with DP arguments, which allow (possessor) extraction in Greek, as illustrated in (10), rather than with adjuncts.

- (7) a. Ti_i ipe [oti efaje t_i]?
 what.ACC said.3SG OTI ate.3SG
 ‘What did she say that she ate?’
- b. Pos_i ipe [oti efaghe to kineziko t_i]?
 how said.3SG OTI ate.3SG the.N.SG.ACC Chinese.N.SG.ACC
 ‘How did she say that she ate the Chinese food?’
- (8) a. Ti_i metanjose [pu efaje t_i]?
 what.ACC regretted.3SG PU ate.3SG
 ‘What did she regret eating?’
- b. *Pos_i metanjose [pu efaje to kineziko t_i]?
 how regretted.3SG PU ate.3SG the.N.SG.ACC Chinese.N.SG.ACC
 ‘How did she regret eating the Chinese food?’

- (9) a. * Pjon_i nevr_iazi o Joryos [otan maloni
 who.ACC get.upset.3SG the.M.SG.NOM George.M.SG.NOM OTAN scold.3SG
 i Maria t_i]?
 the.F.SG.NOM Maria.F.SG.NOM
 Intended: ‘Who does George get upset when Maria scolds?’
- b. * Pos_i nevr_iazi o Joryos [otan maloni i
 how get.upset.3SG the.M.SG.NOM George.M.SG.NOM OTAN scold.3SG the.F.SG.NOM
 Maria ton jo tis t_i]?
 Maria.F.SG.NOM the.M.SG.ACC son.M.SG.ACC 3SG.F.GEN
 Intended: ‘How does George get upset when Maria scolds her son?’
- (10) Pjanu_i δjavase to vivlio t_i?
 whose.GEN read.3SG the.N.SG.ACC book.N.SG.ACC
 ‘Whose did you read the book?’

A reviewer suggests that the extraction evidence is not convincing, since adjunct clauses are transparent for extraction in some languages (Privoznov 2022, Bondarenko 2022). However, the evidence from these languages is limited. The literature shows that only certain adjunct clauses permit extraction, and even then only under restricted conditions, typically with specific extraction types, most notably relativization. This is different from embedded clauses, which freely allow either *wh*-extraction or relativization.

Equally important is the behavior within a given language. In Greek, adjunct clauses uniformly constitute strong islands for extraction, as mentioned already. Against this background, it would be implausible to classify *oti*- or *pu*-clauses as adjuncts solely on the basis that adjunct extraction happens to be marginally possible in some other languages.

In sum, the evidence from clitic doubling, passivization, and extraction converges on the conclusion that *oti*- and *pu*-clauses behave exactly like the corresponding DPs when used as internal arguments. It follows that embedded clauses can occupy the same syntactic position as DPs. Although this view was standard until recently, Bondarenko (2022) denies this possibility, arguing that embedded clauses can never occupy the same position as DPs. Under her view, embedded clauses are adjuncts. Her arguments, however, largely rely on the particular Syntax–Semantics mapping she assumes: because clauses and DPs are semantically composed with verbs in different ways, they must also be merged with verbs from different syntactic positions. As discussed in section 7.2, there is in principle no theoretical reason why this Syntax–Semantics mapping must be correct, and the diagnostics applied in this section in fact strongly support that clauses and DPs occupy the same syntactic position. On this basis, Bondarenko’s position cannot be maintained.

2.2 The internal-external argument asymmetry

Despite being *bona fide* internal arguments, *oti*- and *pu*-clauses are banned from the external argument position, as shown in (11a) and (12a).

- (11) a. * [Oti exis filus] dixni pola ja sena.
 OTI have.2SG friends.M.PL.ACC show.3SG a lot for you.2SG.ACC

- ‘That you have friends shows a lot about you.’
- b. * δixni pola ja sena [oti exis filus].
 show.3SG a lot for you.2SG.ACC OTI have.2SG friends.M.PL.ACC
 ‘It shows a lot about you that you have friends.’
- (12) a. * [Pu exis filus] δixni pola ja sena.
 PU have.2SG friends.M.PL.ACC show.3SG a lot for you.2SG.ACC
 ‘That you have friends shows a lot about you.’
- b. * δixni pola ja sena [pu exis filus].
 show.3SG a lot for you.2SG.ACC PU have.2SG friends.M.PL.ACC
 ‘It shows a lot about you that you have friends.’

Crucially, this is a restriction on the external argument position, not a general ban on clausal subjects. *Oti*-clauses are perfectly grammatical in derived subject positions, as was shown in (6a).

A similar conclusion holds for *pu*-clauses, though a different diagnostic is needed. Because *pu*-clauses are restricted to stative predicates, as we will see in Section 2.3, they cannot be tested with passivization, as stative predicates resist passivization (Landau 2009 i.a.). However, they can appear as arguments of Class III experiencer verbs (e.g., *aresi* ‘like’), which are widely analyzed as unaccusatives (Belletti and Rizzi 1988; Pesetsky 1996; Landau 2009). These verbs select a dative experiencer, optionally realized as a clitic, and a nominative theme that originates internally and can move to subject position. As (13) shows, a nominal theme can appear post-verbally or pre-verbally.

- (13) a. Tu aresi [i Maria].
 3SG.M.DAT like.3SG the.F.SG.NOM Maria.F.SG.NOM
 ‘Maria is appealing to him.’
- b. [I Maria] tu aresi.
 the.F.SG.NOM Maria.F.SG.NOM 3SG.M.DAT like.3SG
 ‘Maria is appealing to him.’

A *pu*-clause can substitute for the theme argument and, crucially, can appear in either the post-verbal (internal argument) position or the pre-verbal (derived subject) position, as in (14).

- (14) a. den tus arese [pu efiyan xoris enimerosi].
 not 3PL.DAT like.3SG PU left.3PL without update
 ‘They did not like that they left without an update.’
- b. [Pu efiyan xoris enimerosi] den tus arese.
 PU left.3PL without update not 3PL.DAT like.3SG
 ‘That they left without an update, they did not like.’

While speakers show a preference for the post-verbal order, the pre-verbal subject position

is grammatical.⁸ The acceptability of (14b) contrasts sharply with the ungrammaticality of (12a). This confirms that *pu*-clauses, like *oti*-clauses, are not barred from subject positions in general, but specifically from the external argument position.

2.3 A novel restriction: stativity in *pu*-complements

The final puzzle comes from a previously unnoticed restriction on *pu*-complements. While *pu* has other functions—introducing adverbial and relative clauses—its use as a complement is unique in requiring a stative matrix predicate.

In its adverbial use, *pu* introduces clauses that express either the reason for an event or the subject matter of an attitude predicate. For instance, it can mark the cause of punishment, (15a), or the topic of an emotional reaction, (15b):

- (15) a. Ton timorisan pu aneptikse afti ti
 3SG.M.ACC punished.3PL PU developed.3SG this.F.SG.ACC the.F.SG.ACC
 siberifora.
 behavior.F.SG.ACC
 ‘They punished him fast because of the fact that he developed this behavior.’
- b. Xarika me ti Maria pu perase tis
 was.happy.3SG with the.F.SG.ACC Maria.F.SG.ACC PU passed.3SG the.F.PL.ACC
 eksetasis.
 exams.F.PL.ACC
 ‘I was happy about Maria for passing the exams.’

Pu also functions as a relative pronoun, introducing restrictive relative clauses:

- (16) a. Anaynorise to pedi pu katestrepse to
 recognized.3SG the.N.SG.ACC kid.N.SG.ACC PU destroyed.3SG the.N.SG.ACC
 poðilato.
 bike.N.SG.ACC
 ‘She recognized the kid who destroyed the bike.’

Finally, *pu* appears as an interrogative pronoun meaning ‘where’:

- (17) a. Pu eftiakse ena spiti o Joryos?
 where built.3SG a.N.SG.ACC house.N.SG.ACC the.M.SG.NOM George.M.SG.NOM
 ‘Where did John build a house?’

What is particularly striking is that the complement use of *pu* differs from these other environments in a crucial way presenting a previously unnoticed asymmetry: *pu*-complement clauses are subject to a stativity requirement on the matrix predicate. That is, only stative predicates can

⁸ A survey of ten native speakers showed a preference for postverbal *pu*-clauses, as in (14a), which received an average rating of 4.6 out of 5. In contrast, preverbal *pu*-clauses, shown in (14b), were rated slightly lower, with an average of 4.2 out of 5.

Two diagnostics will be used to tease apart stative from eventive predicates. The first one is manner adverbial modification. To start with, as illustrated in (18), the eventive, activity, achievement, and accomplishment predicate types within Vendler's (1957) traditional classification permit manner adverbial modification. Conversely, (18a) shows that stative predicates, such as *iche* 'had,' resist modification by manner adverbials. This aligns with the prevailing consensus in the literature that a verb rejecting manner adverbial modification must be stative (Ernst 2016, Maienborn 2005, Rothmayr 2009, among others).⁹ On the other hand, eventive predicates allow for manner adverbial modification, as shown in (18b)–(18d).¹⁰

- I show next that a verb taking a *pu*-clause as its argument must be stative. This is illustrated with *θimono* ‘be/get angry’ and *metanjoni* ‘be/come at a regret state.’ When used intransitively, both verbs can take on an eventive interpretation, as evidenced by the fact that they allow modification with manner adverbs such as *efkola/apotoma* ‘easily/abruptly’ in (19a), (20a). They also allow an eventive usage in (19b) or (20b), where they take a PP-complement introduced by *me* and *ja*, respectively. In (19c) and (20c), *θimono* ‘be/get angry,’ and *metanjoni* ‘be/come at a regret state’ are shown to take a *pu*-clause as an argument. The *pu*-clauses in both cases are thematically interpreted, akin to the corresponding PPs in (19b) and (20b) specifically, as the target of emotion and subject matter, respectively (in terms of Pesetsky 1996). Given this, the contrast

¹⁰ It has been shown that some of the manner adverbs, for example *kala* ‘well’, are often ambiguous between a manner and a degree interpretation. In order to avoid possible confounds that arise with ambiguous adverbs like *kala* ‘well’, I only use adverbs that unambiguously express manner.

in pairs (19b)–(19c) and (20b)–(20c) indicates that although the PPs and *pu*-clauses are thematically interpreted in an identical manner, the matrix verb is obligatorily stative when embedding a *pu*-clause.^{11,12}

- (19) a. *θimose* (apotoma).
was/got.angry.3SG abruptly
'She was/got angry (abruptly).'
- b. *θimose* (apotoma) *me ti* *Maria/* *to*
was/got.angry.3SG abruptly with the.F.SG.ACC Maria.F.SG.ACC/ the.N.SG.ACC
jeghonos.
fact.N.SG.ACC
'She was/got angry (abruptly) with Maria/the fact.'
- c. *θimose* (*apotoma) *pu apoliun* *prosopiko*.
was.angry.3SG abruptly PU fired.3PL personnel.N.SG.ACC
intended: 'She was abruptly angry about the fact that they fired personnel.'
- (20) a. *Metanjose* (efkola).
was/came to be in a state of regret.3SG easily
'She was/came to be in a state of regret easily.'
- b. *Metanjose* (efkola) *ja tin* *epiloji*
was/came to be in a state of regret.3SG easily for the.F.SG.ACC choice.F.SG.ACC
tis na aghorasi mavro *pukamiso*.
her.F.SG.GEN na buy.3SG black.N.SG.ACC shirt.N.SG.ACC
'She was/came to be in a state of regret for her choice to buy a black shirt.'
- c. *Metanjose* (*efkola) *pu aghorase mavro* *pukamiso*.
regretted.3SG easily PU bought.3SG black.N.SG.ACC shirt.N.SG.ACC
intended: 'She was in a state of regret easily that she bought a black shirt.'

¹¹ The meaning of a sentence with an eventive matrix verb can be expressed with a nominalized clause after P:

- (1) *θimose* *apotoma me to oti apoliun* *prosopiko*.
got.angry.3SG abruptly with the OTI fire.3PL personnel.N.SG.ACC
'She got angry abruptly with the fact that they fire personnel.'

¹² For clarity, the verbs *θimose* 'be/get angry' and *metanjose* 'regret' are ambiguous when they are not modified by a manner adverb. When they receive a stative interpretation—for example, when modified by a locative phrase such as *sto parko* 'in the park' (see Alexiadou and Iordăchioaia 2014 for this diagnostic)—they are no longer compatible with manner adverbial modification, as shown below with the verb *metanjose* 'regretted.'

- (1) *Metanjose* *s-to parko* (*efkola).
regretted.3SG in-the.N.SG.ACC park.N.SG.ACC easily
'She regretted easily in the park.'

Oti-clauses exhibit a distinct behavior, as they are not restricted to the argument position of stative predicates. They can function as arguments of stative predicates (e.g., *iksere* ‘know’) which reject manner adverbial modification (21a), or predicates like *sinidhitopii* ‘realize’ or *katalaveni* ‘understand’ that are eventive, as shown by their compatibility with manner adverbial modification (21b).

- (21) a. *Iksere* (**efkola*) *oti* *prepi* *na* *alaksi* *stratijiki*.
 knew.3SG easily OTI must.3SG na change.3SG strategy.F.SG.ACC
 ‘She knew (*easily) that she must change strategy.’
- b. *Arketos* *kosmos* *dhen* *siniðitopii/* *katalaveni* (*efkola*) *oti*
 many.M.SG.NOM people.M.SG.NOM not realize.3SG understand.3SG easily OTI
i *periýrafiki* *kanones* *ine* *ðjaforetiki* *apo*
 the.M.PL.NOM descriptive.M.PL.NOM rules.M.PL.NOM be.3PL different.M.PL.NOM from
tus *riθmistikus*.
 the.M.PL.ACC prescriptive.M.PL.ACC
 ‘People do not understand/realize easily that descriptive rules are different from prescriptive ones.’

The few verbs that accept both *oti*- and *pu*-clauses show the relevant contrast. When combined with *oti*-clauses, these verbs are eventive and therefore allow modification by a manner PP, *me ðiskolia* ‘with difficulty,’ (22a). In contrast, when combined with a *pu*-clause, they are stative and reject modification by the same manner PPs, (22b).

- (22) a. *θimame* *me* *ðiskolia*, *oti* *milises* *s-ti* *Maria*.
 remember.1SG with difficulty OTI talk.2SG to-the.F.SG.ACC Maria.F.SG.ACC
 ‘I remember, with difficulty, that you talked to Mary.’
- b. # *θimame* *me* *ðiskolia*, *pu* *milises* *s-ti* *Maria*.
 remember.1SG with difficulty PU talk.2SG to-the.F.SG.ACC Maria.F.SG.ACC
 ‘I remember, with difficulty, that you talked to Mary.’ Roussou (2019, (7))

The stativity requirement for *pu*-complementation is further confirmed by a second diagnostic: modification by *in*-adverbials (e.g., in five minutes). Such modifiers are known to select for change-of-state/eventive predicates and are incompatible with stative ones (Alexiadou and Iordăchioaia 2014, i.a.). The verb *θimono* ‘get/be angry’ can have an eventive, change-of-state reading, in which case it is compatible with *in*-adverbials, as shown in (23a) and (23b). Yet, when *θimono* embeds a *pu*-clause, this modification becomes ungrammatical (23c), reinforcing the conclusion that the predicate is obligatorily stative in this environment.

- (23) a. *θimose* (*mesa* *se* *pede* *lepta*).
 got.angry.3SG within in five minutes.N.PL.ACC
 ‘She got angry within five minutes.’
- b. *θimose* (*mesa* *se* *pede* *lepta*) *me* *to* *jo*
 got.angry.3SG within in five minutes.N.PL.ACC with the.M.SG.ACC son.M.SG.ACC
tis.
 her.3SG.F.GEN

- ‘She got angry within five minutes with her son.’
- c. θimose (*mesa se pede lepta) pu psifistike o
 got.angry.3SG within in five minutes.N.PL.ACC PU was.voted.3SG the.M.SG.NOM
 nomos.
 law.M.SG.NOM
 ‘She got angry within five minutes that the law was voted.’

In contrast to complement clauses, the stativity restriction vanishes when *pu* is used in other functions. As shown in (24) and (25), matrix verbs are combined with adverbial, relative, or interrogative *pu*-clauses are fully compatible with both manner adverbs and *in*-aspectual modifiers, confirming their eventive nature.

- (24) a. Ton timorisan (yrighora/ se pede lepta) pu iche
 3SG.M.ACC punished.3PL fast in five minutes.N.PL.ACC PU had.3SG
 afti ti siberifora.
 this.F.SG.ACC the.F.SG.ACC behavior.F.SG.ACC
 ‘They punished him fast/in five minutes because of the fact that he had this behavior.’
- b. Tu milise (aschima) pu dhen sevete tus yonis
 3SG.M.DAT talked.3SG badly PU not respect.3SG the.M.PL.ACC parents.M.PL.ACC
 tu.
 his.3SG.M.GEN
 ‘She talked to him badly because of the fact that he does not respect his parents.’
- (25) a. Anaynorise (efkola) (se pede lepta) to pedi pu
 recognized.3SG easily in five minutes.N.PL.ACC the.N.SG.ACC kid.N.SG.ACC PU
 echi to kokino poðilato.
 have.3SG the.N.SG.ACC red.N.SG.ACC bicycle.N.SG.ACC
 ‘She recognized the kid who has the red bicycle fast/in five minutes.’
- b. Pu pije (efkola) (se pede lepta) o
 where went.3SG easily in five minutes.N.PL.ACC the.M.SG.NOM
 Joryos?
 George.M.SG.NOM
 ‘Where did John go fast/easily?’

3 Analysis

To resolve these puzzles, I propose an analysis that reverses the traditional direction of selection. I argue that *oti* and *pu* select a noun. In complement clauses, this is a light noun that must be licensed via incorporation into the matrix verb. This single mechanism, combined with a semantic distinction between meaning of these clauses, accounts for both the internal-external argument asymmetry and for their near-complementary distribution.

3.1 Reversed selection and the internal-external argument asymmetry

I partly adopt Arsenijević (2009) in assuming that embedded clauses are built on a structure in which C selects a light noun in its specifier (see also Moltmann 2019, 2024). Concretely, *oti* and *pu* bear an uninterpretable [n]-feature (see Adger 2003; Collins and Stabler 2016; Zeijlstra 2020; Chomsky 1995; Merchant 2019 for analyses of selection in terms of feature checking), which is checked by merging a light noun (n) in their specifier.

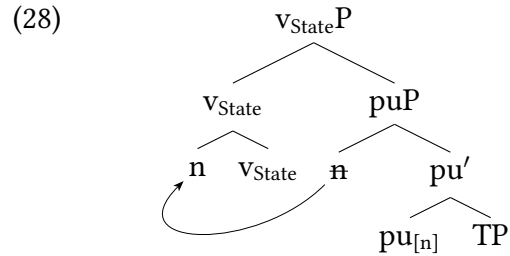
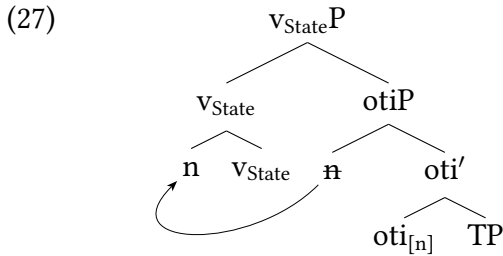
In Arsenijević’s analysis, this silent noun also undergoes a relativization step, which the present proposal does not adopt. Specifically, he assumes that the noun is externally merged in Spec,ForceP, moves to Spec,CP, and is subsequently deleted through matching with a higher noun. Because the clause is analyzed as a relative clause, the structure also includes an additional DP layer, as illustrated below. The higher noun then incorporates into the matrix verb.

$$(26) \quad [\text{DP D } [\text{NP claim } [\text{CP claim } [\text{C}' \text{ C } [\text{ForceP claim } [\text{Force}' \text{ Force } [\text{IP John kissed Mary}]]]]]]]$$

Under this structure, the embedded clause, the CP, is not an argument of the matrix verb, since it is embedded under a DP and NP layer. The analysis proposed here departs from Arsenijević in not adopting the relativization component. As discussed in the previous section, this component is not motivated because embedded clauses in Greek do not display the distribution or interpretation of DPs. I therefore retain only the idea that a noun merges in the specifier of the complementizer and subsequently incorporates into the matrix verb. Without the DP layer, the CP is merged directly in the verb’s complement position and thus functions as its argument.

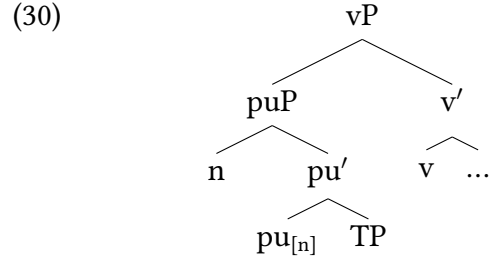
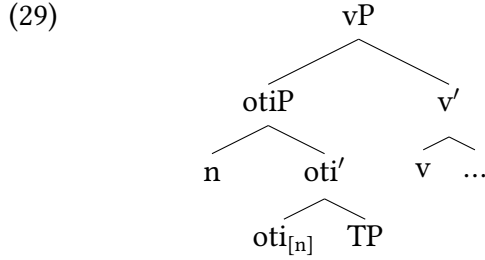
I begin by showing how this analysis works when the clause occupies the internal argument position of the verb.

When an *oti*/*pu*-clause is an internal argument, it is merged as the complement to an aspectual head, v_{State} or v_{Event} (see Borer 2005; Merchant 2019 for the idea that internal arguments are introduced by aspectual heads). The light noun in Spec,*oti*P or *pu*P can successfully incorporate into v , satisfying its licensing requirements.



When *oti*- and *pu*-clauses appear as derived subjects of passivized verbs, they are built from the same structures. Incorporation of the silent noun takes place before the clause moves to subject position.

This analysis directly accounts for the absence of external arguments. The external argument is externally merged in Spec,vP (29)–(30). From this specifier position, the light noun cannot incorporate into a higher head, as the closest potential host—T—is a functional head and therefore unable to license a light noun. Licensing requires incorporation into a lexical v head (e.g., *make*, *do*, *be*), rather than into a functional head such as T.



This proposal also correctly predicts that bare *oti*/*pu*-clauses are ungrammatical as complements of Ps. Like T, P cannot serve as a host for noun incorporation (Hale and Keyser 1993). This prediction is borne out: both *oti*- and *pu*-clauses are excluded after Ps. Consider the data in (31)–(32). The verbs *djafono* ‘disagree’ and *metanjono* ‘regret’ can take either a PP or a bare *oti*/*pu*-clause as their complement (31b, 32b). However, when these verbs select a PP, the embedded clause cannot combine with the P, as shown by the ungrammaticality of (31c) and (32c).

- (31) a. *djafono me kati.*
disagree.1SG with something.N.SG.ACC
‘I disagree with something.’
- b. *djafono oti prepi na kanume ependisis.*
disagree.1SG OTI must.3SG na do.1PL investments.F.PL.ACC
‘I disagree that we should invest.’
- c. **djafono me oti prepi na kanume ependisis.*
disagree.1SG with OTI must.3SG na do.1PL investments.F.PL.ACC
intended: ‘I disagree with the idea that we should invest.’
- (32) a. *Metanjono ja kati.*
regret.1SG for something.N.SG.ACC
‘I regret for something.’
- b. *Metanjono pu paremina.*
regret.1SG PU stayed.1SG
‘I regret that I stayed.’
- c. **Metanjono ja pu paremina.*
regret.1SG for PU stayed.1SG
intended: ‘I regret for staying.’

In sum, the light-noun analysis captures both the internal–external argument asymmetry and the restriction against *oti*- and *pu*-clauses after Ps.¹³ What remains is the question of their near-complementary distribution after verbs (see 1), which I turn to in the next section.

¹³ An anonymous reviewer asks whether examples like (12) are ungrammatical not because of the syntax of *pu*-clauses, but rather due to a semantic restriction—specifically, that situations, which *pu*-clauses denote (as discussed in Section 3.2), cannot serve as subjects of verbs like *dhihni* ‘show.’ However, this alternative explanation is untenable. Greek readily allows situation-denoting nouns, such as *katastasi* ‘situation,’ to occur as subjects of *dhihni* ‘show,’ as shown below. This observation further supports the view that *pu*-clauses are syntactically constrained from appearing in certain subject positions.

3.2 The semantics of selection: content vs. situation

Having accounted for the internal-external argument asymmetry, we turn to the near-complementary distribution of *oti* and *pu*. I argue this reflects a fundamental semantic dichotomy between content and situation (Bondarenko 2022; Moltmann 2021), which applies to the clauses themselves and the nouns they combine with.

We begin with nouns. Content vs situation nouns are distinguished by predicates like *alithis* ‘true’ or *lanthasmeni* ‘mistaken,’ which can combine only with content nouns, (33), and verbs like *simveni* ‘happen,’ which can combine only with situation nouns, (34).

- (33) a. Afti i fimi ine aliθis/
 this.F.SG.NOM the.F.SG.NOM rumor.F.SG.NOM be.3SG true.F.SG.NOM
 lanθasmeni.
 mistaken.F.SG.NOM
 ‘This rumor is true/mistaken.’
- b. # Afti i katastasi ine aliθis/
 this.F.SG.NOM the.F.SG.NOM situation.F.SG.NOM be.3SG true.F.SG.NOM
 lanθasmeni.
 mistaken.F.SG.NOM
 ‘This situation is true/mistaken.’
- (34) a. # Tetjes fimes/ iðees/ proiðopiisis/
 such.F.PL.NOM rumors.F.PL.NOM/ ideas.F.PL.NOM/ warnings.F.PL.NOM/
 pepiθisis/ ipothesis/ theories ðen simvenun sixna.
 beliefs.F.PL.NOM/ hypotheses.F.PL.NOM/ theories.F.PL.NOM not occur.3PL often
 ‘Such rumors/ideas/warnings/beliefs/hypotheses/theories do not happen often.’
- b. Tetjes periptosis/ katastasis ðen simvenun sixna.
 such.F.PL.NOM situations.F.PL.NOM/ cases.F.PL.NOM not occur.3PL often
 ‘Such situations/cases do not happen often.’

Two facts suggest that the content vs situation distinction is important for the distribution of *oti* vs *pu*.

First, *oti*-clauses can only combine with content nouns whereas *pu*-clauses can only combine with situation nouns:

- (35) a. Akuse ti fimi/ iðea/ neo/
 heard.3SG the.F.SG.ACC rumor.F.SG.ACC/ idea.F.SG.ACC/ news.N.SG.ACC/
 jeyonos oti/*pu i Ji ine strogili.
 fact.N.SG.ACC OTI/*PU the.F.SG.NOM earth.F.SG.NOM be.3SG round.F.SG.NOM
 ‘She heard the rumor/idea/news/fact that the earth is round.’

(1) Mía tetjia katastasi pu o pateras ine apon dhichni pola.
 a such situation PU the father be.3SG absent show.3SG a lot
 ‘Such a situation where the father is absent shows a lot.’

- b. Afora tin periptosi/ peristasi/ katastasi
 concern.3SG the.F.SG.ACC case.F.SG.ACC/ circumstance.F.SG.ACC/ situation.F.SG.ACC
 pu/*oti o pateras ine apon.
 PU/*OTI the.M.SG.NOM father.M.SG.NOM be.3SG absent.M.SG.NOM
 ‘It concerns the case/circumstance/situation that the father is absent.’

Second, when nominalized, verbs that combine with *oti* (e.g., *believe*) behave like *rumor*: they are compatible with *true/mistaken* but not with *happen*, (36). In contrast, when nominalized, verbs that combine with *pu*-clauses pattern (e.g., *be saddened*) like *situation*-nouns: they are incompatible with *true/mistaken* but are compatible with *happen*, (37).

- (36) a. I pisti oti i maskes ine pali
 the.F.SG.NOM belief.F.SG.NOM OTI the.F.PL.NOM masks.F.PL.NOM be.3PL again
 ipoxreotikes ine aliθis/ lanθasmeni.
 mandatory.F.PL.NOM be.3SG true.F.SG.NOM mistaken.F.SG.NOM
 ‘The belief that masks are obligatory again is true/mistaken.’
- b. # An afti i pisti mu iche simvi
 if this.F.SG.NOM the.F.SG.NOM belief.F.SG.NOM me.1SG.DAT had.3SG happened
 s-ta dekaeksi mu, θa imun alos
 in-the sixteen.N.PL.ACC mine.1SG.GEN would be.1SG different.M.SG.NOM
 anθropos.
 person.M.SG.NOM
 ‘If this belief had happened to me when I was sixteen, I would be a different person.’
- (37) a. # I stenaxorja tus pu efije i
 the.F.SG.NOM sadness.F.SG.NOM their.3PL.GEN PU left.3SG the.F.SG.NOM
 Maria ine aliθis/ lanθasmeni.
 Maria.F.SG.NOM be.3SG true.F.SG.NOM mistaken.F.SG.NOM
 ‘Their sadness that Maria left is true/mistaken.’
- b. ? An afti i stenaxoria mu iche simvi
 if this.F.SG.NOM the.F.SG.NOM sadness.F.SG.NOM me.1SG.DAT had.3SG happened
 s-ta dekaeksi mu, θa imun alos
 in-the sixteen.N.PL.ACC mine.1SG.GEN would be.1SG different.M.SG.NOM
 anθropos.
 person.M.SG.NOM
 ‘If this sadness had happened to me when I was sixteen, I would be a different person.’

The generalization is thus straightforward: only content-denoting nouns and verbs combine with *oti*, while only situation-denoting ones combine with *pu*. I propose that this follows if *oti*-clauses themselves denote content, while *pu*-clauses denote situations.¹⁴ The compatibility

¹⁴ As noted by an reviewer, this distinction predicts that verbs like *occur* and *happen* should not be able to combine

requirement arises because only clauses and lexical items with matching meanings can combine, content-denoting clauses can only be combined with content-denoting nouns and situation-denoting clauses can only be combined with situation-denoting nouns.¹⁵ Specifically, this requirement ensures that only a content-denoting nominal, which is subsequently verbalized, can be merged in the specifier of *oti*. Likewise, only a situation-denoting nominal can be merged in the specifier of *pu*. This analysis directly accounts for the data in (35), (36), and (37).

The idea that *oti*-clauses denote content whereas *pu*-clauses denote situations is further supported by factivity entailments. As argued by Bondarenko (2022), clauses denoting situations are necessarily evaluated in the real world due to their internal composition, giving rise to factivity. In contrast, clauses denoting content need not be factive. This contrast holds in Greek: *pu*-clauses, expressing situations, are factive, whereas *oti*-clauses are not.¹⁶ The following examples illustrate this:¹⁷

with *oti*-clauses because they describe content. Additionally, *pu*-clauses should also be blocked, as these verbs are change-of-state or eventive, while *pu*-clauses can only be combined with stative predicates. Indeed, this prediction is borne out as shown below:

- (1) *Ihe simvi OTI/PU kanume lathos.
 had happened.3SG OTI/PU make.1PL mistake.N.SG.ACC
 ‘It had happened that we make a mistake.’

These predicates can only take embedded clauses introduced with another complementizer, not discussed here, namely, *na*. This element is used in Greek where other languages use the subjunctive or infinitives (see Terzi 1992 i.a.).

¹⁵ In Bondarenko (2022), this compatibility arises as a result of the way Predicate Modification, the operation that is responsible for the semantic operation via which nouns and clauses are combined, works.

¹⁶ Just like *thimude* ‘remember’ in (38), there are around three more verbs that can be combined both with *oti*- and *pu*-clauses. For these verbs, it is important to note that they are ambiguous. This ambiguity can be seen more clearly with verbs like *vlepo*, which can be used as an attitude verb, meaning ‘understand,’ or as a pure perception verb, meaning ‘see.’ For these types of verbs, we can hypothesize that they are derived from two distinct homophonous roots. As expected, the root that gives rise to the attitude meaning can be combined with an *oti*-clause, while the perception meaning arises when this root is combined with a *pu*-clause. A reviewer asked about the two possible interpretations of *thimude* ‘remember.’ As noted by Roussou (2019) and Christidis (1982), *thimude* has a direct perception interpretation (i.e., an immediate recollection of an event) when it combines with *pu*. In contrast, when it appears with an *oti*-clause, this reading is unavailable, and the verb instead takes on a different meaning, closer to *infer*.

¹⁷ An anonymous reviewer suggested an alternative analysis that would treat *pu* and *oti* as allomorphs of a single functional head, whose realization is conditioned by the selecting verb. While such an analysis is plausible at first glance, it rests on the crucial assumption that *oti* and *pu* are realizations of the same underlying feature bundle. The empirical data, however, strongly indicates that they are fundamentally different items. A purely allomorphic account cannot explain the systematic differences between them, which span from their core meaning to their syntactic behavior. Semantically, the two complementizers introduce different kinds of objects: *oti*-clauses express propositional content, whereas *pu*-clauses denote situations. This distinction shows up most notably in factivity, as *pu*-clauses carry a presupposition that their complement is true, while *oti*-clauses do not. Their syntax differs just as significantly. *Pu*-clauses can be nominalized and appear in relative or interrogative clauses, environments where *oti*-clauses are prohibited (see also Roussou 2000 for differences between the two). These differences are not expected if *oti* and *pu* are different spell-outs of an identical head.

- (38) a. θ imude oti i yata efaje to psari,
remember.3PL OTI the.F.SG.NOM cat.F.SG.NOM ate.3SG the.N.SG.ACC fish.N.SG.ACC
ala i yata dhen to iche fai s-tin
but the.F.SG.NOM cat.F.SG.NOM not 3SG.N.ACC had.3SG eaten in-the.F.SG.ACC
praymatikotita.
reality.F.SG.ACC
‘They remember that the cat had eaten the fish but the cat had actually not eaten it.’
- b. θ imude pu i yata efaje to psari, # ala
remember.3PL PU the.F.SG.NOM cat.F.SG.NOM ate.3SG the.N.SG.ACC fish but
i yata dhen to iche fai s-tin praymatikotita.
the.F.SG.NOM cat not 3SG.N.ACC had eaten in-the reality.F.SG.ACC
‘They remember that the cat had eaten the fish but the cat had actually not eaten it.’

4 Deriving the argument/adjunct divide

The idea that *pu* projects both a complement and a specifier can now be extended to solve the stativity puzzle. The key lies in the syntactic difference between how arguments and adjuncts are licensed.

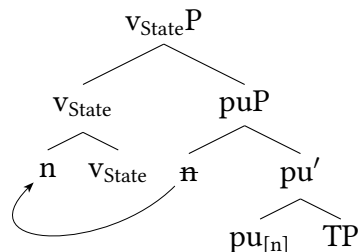
4.1 The locus of stativity: aspectual heads

As previously discussed, I assume that internal arguments are introduced in the complement position of aspectual heads, namely v_{State} or v_{Event} . The stativity restriction on *pu*-complements can thus be attributed to the selectional properties of these heads: only v_{State} can select a *puP* as its complement, whereas v_{Event} cannot.¹⁸ In contrast, both v_{State} and v_{Event} may select an *otiP* as their complement. Selectional Locality (Sportiche 2005 i.a.) is satisfied in this configuration because, as shown in the structure below (repeated from 28), *puP* is merged in the complement position of v_{State} , thereby allowing the licensing of the light noun.¹⁹

¹⁸ Assuming that content-denoting and situation-denoting clauses — that is, *oti*- and *pu*-clauses respectively — have distinct syntactic categories, as in Bondarenko (2022), it is the category of each clause that is selected.

¹⁹ Admittedly, the stativity requirement, as derived from the selectional properties of aspectual heads, is not yet fully explained in the current proposal. The analysis does not, for example, offer a principled reason for the inability of v_{Event} to select a *pu*-clause; this restriction is simply encoded as a selectional feature. An advantage of this proposal, however, is that it accounts for the absence of the stativity requirement in adjunct clauses. Under a semantic analysis in which *pu* imposes a presupposition/definedness condition, such that it is true only of relations between stative events and the situations described by the embedded TP, the contrast would remain unexplained. Such an analysis would be just more stipulative than a selectional account, as it would stipulate an arbitrary denotation, and would additionally have to stipulate that the presupposition arises only in complement clauses, since the stativity requirement is not observed in adjunct clauses.

(39)



This analysis correctly predicts that *pu*-clauses cannot function as arguments of eventive verbs, whereas *oti*-clauses may serve as complements to either stative or eventive predicates.

4.2 Licensing adjuncts: selection from the outside-in

The stativity restriction disappears with adverbial and relative *pu*-clauses (see Section 2.3) because adjuncts are licensed differently. Following a growing body of work (Bruening 2013; Hunter 2015; Zeijlstra 2020; Hewett 2023; Neeleman et al. 2023, i.a.), I assume that while heads select their arguments, adjuncts select their hosts. An adjunct *pu*-clause is merged higher in the structure (e.g., adjoined to $v_{\text{State}}P$) and cannot be selected by an aspectual head because it is not in a local head-complement relation with it. As a result, it is exempt from the selectional requirements of such a head.

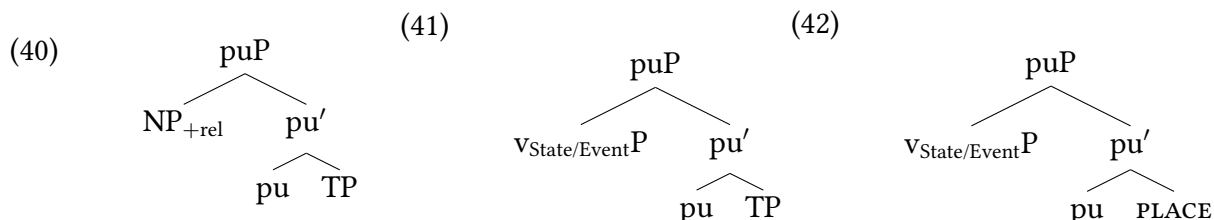
To illustrate the range of structural positions that *pu* can occupy, let us consider its various uses. Crucially, across these uses, *pu* projects the same underlying configuration, comprising both a complement and a specifier. However, I assume that *pu* has flexible selectional requirements, allowing different elements—beyond a light noun and a TP—to merge in these positions.

In relative clauses, *pu* takes a TP complement and a full NP bearing a [+relative] feature in its specifier (40). In this configuration, a D head merges above the *puP*, as is standardly assumed (Kayne 1994 i.a.), yielding a relative clause.

In its adverbial adjunct use, *pu* likewise takes a TP complement and selects the verbal projection it modifies, $v_{\text{State/Event}}P$, (41).

Finally, in its interrogative use, *pu* selects a silent PLACE noun, which gives rise to its locative interpretation, and its host, i.e., $v_{\text{State/Event}}P$ (42).

To summarize, *pu* can select either (i) a TP and a noun (with or without a [+relative] feature), (ii) a TP and a $v_{\text{State/Event}}P$, or (iii) a $v_{\text{State/Event}}P$ and a PLACE nominal.



In all these configurations, *puP* is not merged in the complement position of an aspectual head, so it cannot be subject to the selectional requirements of these heads. This accounts for the

absence of the stativity restriction in the uses of *in* relative clauses and the rest of *pu*'s adjunct uses.²⁰

In sum, *pu*-clauses can function both as arguments and as adjuncts. The unifying mechanism underlying *pu*'s distributional flexibility is selection, formalized as feature-checking within a feature-driven Merge framework. This approach extends to adjuncts, with a crucial difference in the directionality of selection: heads select their arguments, whereas adjuncts select their hosts. Complement *pu*-clauses are subject to the stativity requirement because they are merged in the complement position of aspectual heads, which allows the licensing of the silent noun in the specifier of *pu*. On the other hand, adjunct clauses escape this requirement because they are merged higher in the structure, above these heads.²¹

5 Nominalized clauses

As we have seen, bare *oti*-clauses are excluded from both the external argument position and the complement position of prepositions. However, these restrictions are lifted when the clause is nominalized with the definite article *to* 'the.' Specifically, *oti*-clauses may appear as external

²⁰ An anonymous reviewer also asks why the restriction observed with complement *pu*-clauses—namely, that they can only combine with situation nouns—does not extend to relative clauses, as in (40). As discussed in Section 3.2, this restriction follows from the semantics of *pu*-clauses, which denote situations and therefore can only combine with nouns of similar meaning. The straightforward answer is that the situation meaning simply does not arise in relative clauses, or in other uses of *pu*. Consequently, when *pu* occurs in a relative clause modifying a noun, it can combine with any type of noun.

The question, then, is where the situation meaning originates. Since it cannot come from *pu* itself, it must be introduced by the structure that *pu* combines with. Following Bondarenko (2022), this meaning arises through a projection in the left periphery, namely *sitP*. Thus, in complement clauses, the complement of *pu* is a *sitP* rather than a *TP*. This structural difference gives rise to the situation interpretation of *pu*-clauses, which in turn explains why they can only combine with situation nouns. In relative clauses, by contrast, the complement of *pu* is a plain *TP*; since the situation meaning is absent, there is no restriction on the type of noun it can modify.

²¹ The requirement for stative predicates cannot be attributed to the factive interpretation of *pu*-clauses. This is because factive DP arguments, such as *to jeghonos* 'the fact' can be used with eventive predicates, as shown by the compatibility of the verb with a manner adverb:

- (1) Praymati, ton enoxli (efkola) afto to jeyonos_i.
indeed 3SG.M.ACC annoy.3SG easily this.N.SG.ACC the.N.SG.ACC fact.N.SG.ACC
'Indeed, this fact annoys him easily.'

Similarly, the stativity requirement cannot be attributed to a potential locative interpretation of *pu*. As illustrated in (25b), when *pu* is used as an interrogative expression meaning *where*, the verb combined with *pu* can indeed be eventive. Lastly, the stativity requirement does not seem to arise from the situation meaning of *pu*-clauses. As shown, situations can be arguments of eventive predicates:

- (2) Adilifθike (γriyora) (se pede lepta) tin katastasi.
realized.3SG fast in five minutes the.F.SG.ACC situation.F.SG.ACC
'She realized the situation fast/in five minutes.'

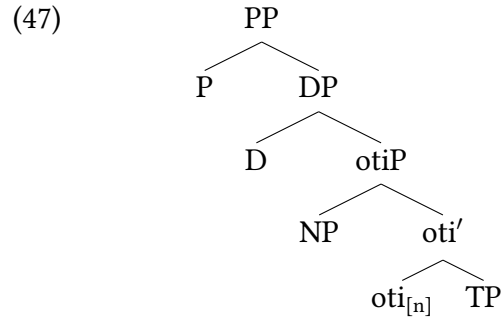
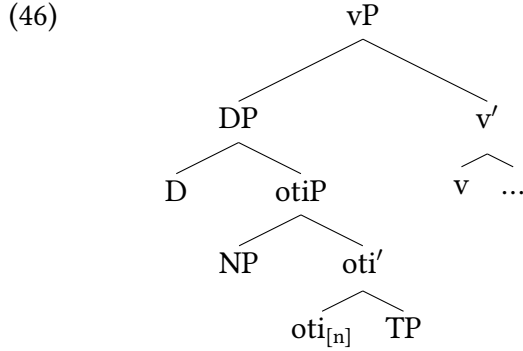
arguments and as complements of Ps, provided they are nominalized with *to* ‘the,’ as illustrated in (43) and (45a) (Roussou 1991; Faure 2022). By contrast, *pu*-clauses resist nominalization altogether, as shown in (44) and (45b).

- (43) a. *(To) *oti echis filus dixni pola ja sena.*
the OTI have.2SG friends.M.PL.ACC show.3SG a lot for you.2SG.ACC
‘That you have friends shows a lot about you.’
b. *dixni pola ja sena *(to) oti echis filus.*
show.3SG a lot for you.2SG.ACC the OTI have.2SG friends.M.PL.ACC
‘It shows a lot about you that you have friends.’
- (44) a. *To *pu echis filus dixni pola ja sena.*
the PU have.2SG friends.M.PL.ACC show.3SG a lot for you.2SG.DAT
‘That you have friends shows a lot about you.’
b. **dixni pola ja sena to pu echis filus.*
show.3SG a lot for you.2SG.DAT the PU have.2SG friends.M.PL.ACC
‘It shows a lot about you that you have friends.’
- (45) a. *Fenete apo *(to) oti ekleyan.*
become.obvious.3SG from the OTI was.crying.3PL
‘It becomes obvious from the fact that they were crying.’
b. **Fenete apo to pu ekleyan.*
become.obvious.3SG from the PU was.crying.3PL
‘It becomes obvious from the fact that they were crying.’

Nominalized clauses have been analyzed in two main ways. On Roussou’s (1991) account, they instantiate a [_{DP} D CP] structure, with D directly merged with the CP realized by the *oti*-clause.²² By contrast, Moulton (2020) proposes a structure [_{DP} D [_{NP} NP CP]], where a silent content noun first merges with the CP, forming an NP that is subsequently merged with D.

Following Moulton (2020), I analyze these as DPs containing a silent noun: [_{DP} D [_{NP} NP CP]]. In this structure, the element that satisfies the [n]-feature of *oti* is not a light noun requiring incorporation, but a full (albeit silent) noun such as FACT or some other content noun such as *rumor* (see Angelopoulos et al. 2025 for additional evidence on a silent noun in nominalized clauses). This noun does not undergo incorporation. This accounts straightforwardly for why nominalization rescues these otherwise illicit configurations. The two structures below illustrate *oti*-nominalized clauses occupying the external argument position of the P-complement position.

²² Roussou (1991) makes somewhat conflicting claims about *to*. On the one hand, it is treated as an expletive D required for case; on the other, it is analyzed as a contentful definite determiner incompatible with *pu*, which is itself a definite element.



A question that arises is why the definite article must appear when this noun is present in Spec,otiP. To understand this, it is important to consider the corresponding cases where an *oti*-clause combines with an overt content noun.²³ Examining such cases, we find that the noun cannot appear bare. Instead, as shown in (48), a definite D must be merged with the noun, particularly when these structures occur in the external argument position or as complements of P. Given this, the definite D *to* ‘the’ is likewise required when a silent NP is merged with *oti*, as in (46) and (47). Since the definite D is needed whenever a full NP merges in *oti*’s specifier, this explains why bare *oti*-clauses cannot be licensed in the external argument position.²⁴

- (48) a. * (I) fimi oti tha apolithun tromazi tus erghazomenus.
 the rumor OTI will be.fired.3PL scare.3SG the employees
 ‘The rumor that they would get fired scares the employees.’
 b. Akuse ja *(ti) fimi.
 heard.3SG for the rumor
 ‘She heard about the rumor.’

²³ As in Arsenijević (2009), it may be that the noun moves out of the specifier, as illustrated below. This movement step is often assumed in analyses of relative clauses as well, where the NP is moved from Spec,CP, creating an NP layer.

(1) [DP D [NP NP [otiP NP [oti' oti TP]]]]

²⁴ An anonymous reviewer raises the possibility that *oti*-clauses are nPs headed by a silent noun. Since *oti*-clauses denote content, they would require this silent noun to be a content noun. However, content nouns cannot appear bare in the complement position of clause-embedding verbs, as shown below, just as in the external argument position and after Ps. This suggests that if they were nPs, they would require an overt determiner, thus yielding a nominalized clause (a DP), contrary to what is observed, rather than a bare *oti*-clause.

(1) Kseri *(tin) fimi.
 know.3SG the.F.SG.ACC rumor.F.SG.ACC
 ‘He knows the rumor.’

Likewise, the sentence above means that if a full silent noun is merged in *oti*’s specifier—as we have assumed for the external argument position and after Ps—a D would still be needed, giving rise to a nominalized clause in the internal argument position as well.

The inability of *pu*-clauses to nominalize also follows. As established, *pu* can only combine with a situation noun. If we assume that Greek lacks a silent situation noun, then no suitable nominal element is available to satisfy *pu*'s selectional feature in a nominalized structure. Consequently, *to pu*-clauses are ungrammatical.²⁵

6 Alternative analyses

In so far we have seen that the proposed analysis offers an account of several complex patterns in Greek. It is worth briefly considering why other theories of clausal complementation fail to do so. We will focus in particular on the external-internal argument asymmetry and the restriction after *Ps*.

First, accounts that posit a categorial restriction on the subject position—such as Hartman (2012), who argues that Spec,TP can only host DPs—run into problems. Greek provides independent evidence for an alternative subject position besides Spec,TP: DP subjects can surface postverbally in Spec,vP, with the verb moving to T (Alexiadou and Anagnostopoulou 2001 i.a.). If so, one might expect clausal subjects to be licensed in Spec,vP, which is not restricted to DPs. Yet this expectation is not borne out: *oti*-/*pu*-clauses cannot appear postverbally either, as shown in (11)–(12).

Second, Case-based theories such as Stowell's (1981) Case Resistance Principle predict that clauses functioning as external arguments should be licit under two conditions: (a) if they move out of the Case-marked subject position (e.g., to a topic position), or (b) if they are base-generated in such a position and resumed by a silent pronoun, as illustrated in (49) (Koster 1978). While this may capture the availability of sentences in English (e.g., such as *That he has blood in his hands proves his complicity*), it incorrectly predicts that the Greek sentences in (11)–(12) should be recoverable via topicalization, contrary to fact:

- (49) [TopicP [otip ...]_i [Topic' Topic [... [vP pro_i ...]]]]

Third, Manzini and Savoia's (2018) Agree-based account fares no better. Under this view, clausal subjects lack phi-features and so they cannot enter an Agree relation with *v* or *T*. This analysis predicts that clausal subjects should never be possible—again at odds with the Greek data, where, as shown in (6a), can serve as subjects.

Approaches that assume embedded clauses are merged in a position distinct from DP arguments are challenged by robust Greek diagnostics for argumenthood, including clitic doubling, passivization, and extraction. Moreover, Bondarenko (2022) argues that only DPs can occupy the external-argument position. This analysis predicts that apparent clausal external arguments should be possible if they are base-generated in a high Topic position, with a silent DP operator

²⁵ Angelopoulos et al. (2025) argue that this correlates with the internal structure of situation nouns in Greek. In contrast to content nouns like *fact*, situation appear to have more internal structure, which prevents them from being realized silently. The underlying idea is that a noun can remain unpronounced only if it corresponds to a minimal structural layer.

or *pro* in Spec,vP satisfying the DP requirement, as in (49) (see Anagnostopoulou 1997 for the availability of such structures in Greek).²⁶

Finally, a broader difficulty for these approaches is the post-P restriction. Few proposals account for it directly such as Stowell’s (1981) Case Resistance Principle but, as we saw, this cannot be adopted because in contrast to the proposed analysis, it cannot predict the distribution of *oti*-/ *pu*-clauses in the external argument position. A reviewer suggests that Ps semantically select individuals rather than propositions, blocking clausal complements. While possible in principle, this relies on the assumption that all Ps are semantically contentful. Yet Greek functional Ps have been argued to be semantically vacuous (Angelopoulos et al., 2020), making a purely semantic explanation for the restriction unlikely.

7 Broader implications

The proposed analysis of clausal complementation has broader consequences, including cross-linguistic extensions, the role of selection, and the Syntax-Semantics interface.

7.1 Cross-linguistic extensions

The proposal that adjuncts select their hosts can be extended to explain distributional puzzles raised by embedded clauses in other languages. Here we focus on embedded clauses of Uyghur introduced by the *say*-element *dep*, (50).

- (50) Mahinur Tursun-(ni) göshnan-ni et-t-i de-p oyla-y-du.
 Mahinur Tursun-ACC meatbread-ACC make-PST-3 say-CNV think-NONPST-3
 ‘Mahinur thinks that Tursun made meatbread.’ (Major 2024, (2))

Major (2024) argues that the marker *dep*, which introduces embedded clauses in Uyghur, is decomposable into two morphemes: the verb *de* ‘say’ and the converb *-ip*. On this view, *de* heads a VP, which is merged as the complement of *-ip*, yielding the structure in (51).

- (51) [_{ConverbP} *ip*_{Converb} [_{VP} *de*_V ...]]

As Major observes, *dep*-clauses exhibit a striking distribution: they can attach as adjuncts, either to TP or to VP, but crucially, they cannot appear in argument positions—neither as external nor internal arguments. The external argument restriction is shown in (52), where a *dep*-clause cannot serve as the subject of a predicate like ‘surprise.’

²⁶ A reviewer points out that the pronoun in the external argument position should also be of the same semantic type as the clause, namely a predicate of individuals with prepositional content. On this view, the structure should be ruled out because an external argument must denote an individual (type *e*). This alternative, however, does not go through. The pronoun occupying the external argument position would move from Spec,vP to Spec,TP and would be converted into an individual-denoting expression via Trace Conversion (Fox 1999). Therefore, there is no issue with the semantic composition of such a structure.

- (52) * Tursun-(ni) ket-t-i de-p (xewer) méni hayran qal-dur-d-i.
 Tursun-ACC leave-PST-3 say-CNV news 1SG.ACC surprise remain-CAUS-PST-3
 Intended: ‘(The news) that Tursun left surprised me.’ (Major 2024, (49b))

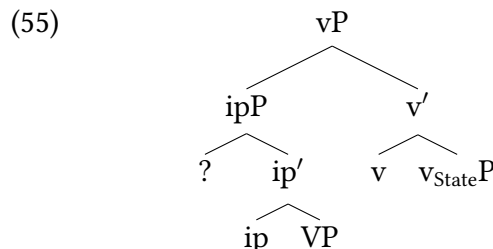
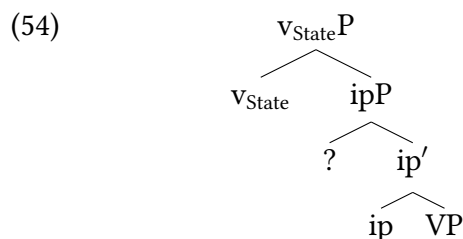
Turning to internal arguments, consider the intransitive verb *warqiri-d-i* ‘screamed,’ which does not normally take a complement. In combination with a *dep*-clause, however, the verb is coerced into a speech predicate. Thus, in (53) the matrix clause means simply ‘Mahinur screamed,’ while the *dep*-clause contributes the propositional content of the scream.

- (53) Mahinur [Tursun-ni ket-t-i de-p] warqiri-d-i.
 Mahinur Tursun-ACC leave-PST-3 say-CNV scream-PST-3
 ‘Mahinur screamed, saying that Tursun left.’ (Major 2024, (38))

These facts follow naturally if adjuncts select their hosts—either TP or $v_{\text{State/Event}}P$. On this view, Uyghur *dep*-clauses have two core properties: (1) they can merge at either the $v_{\text{State/Event}}P$ or TP level, and (2) they are excluded from argument positions.

The first property derives from the selectional profile of *-ip*. Like *pu*, *-ip* selects both its immediate VP complement (headed by *de*) and a larger clausal host ($v_{\text{State/Event}}P$ or TP). This dual selection accounts for the two possible attachment sites.

The restriction against argument positions follows as well. If the ConverbP is merged as an internal argument, as in (54), it is too low to satisfy *-ip*’s selectional requirement for $v_{\text{State/Event}}P$ or TP. If it is merged as an external argument, as in (55), it again fails to meet these requirements, since the necessary Head–Complement or Spec–Head relation with $v_{\text{State/Event}}P$ /TP does not obtain.²⁷



7.2 The role of selection in clausal complementation

Our findings suggest that selection is invariably involved in clausal complementation. This follows from the view that selection serves as a feature-checking mechanism underlying both argumenthood and adjunction: heads select their arguments, and adjuncts select their hosts. A direct

²⁷ An alternative account is that *dep*-clauses are simply not selected by predicates (see Bochnak and Hanink 2022 on all arguments). On this view, since transitive predicates select DPs, *dep*-clauses are excluded from argument positions. While plausible, this approach cannot fully rule out cases where an ipP is externally merged in a high (e.g., Topic) position with a DP operator filling the argument slot. By contrast, the adjunct-selection analysis rules these out straightforwardly, since *-ip*’s selectional requirements cannot be met if ipP is merged in Spec,TopicP.

consequence of this analysis is that selection is necessarily involved in the licensing of embedded clauses, regardless of whether they function as arguments or adjuncts.

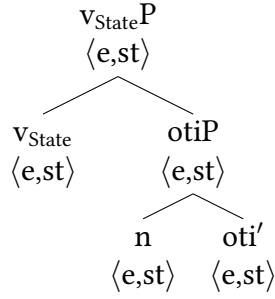
This approach challenges the conclusion of Bochnak and Hanink (2022), who, based on evidence from Washo, argue for a more restricted view in which selection is limited to argument clauses. The key point of divergence lies in the analysis of adjuncts. Bochnak and Hanink’s conclusion holds only if adjuncts neither select nor are selected. However, if adjuncts themselves act as probes that select their hosts, as assumed here, then selection remains an indispensable mechanism in the licensing of clauses. This, in turn, leads to the broader theoretical conclusion that clausal selection and modification are not mutually exclusive.

7.3 The semantics of embedded clauses: insights into the syntax–semantics interface

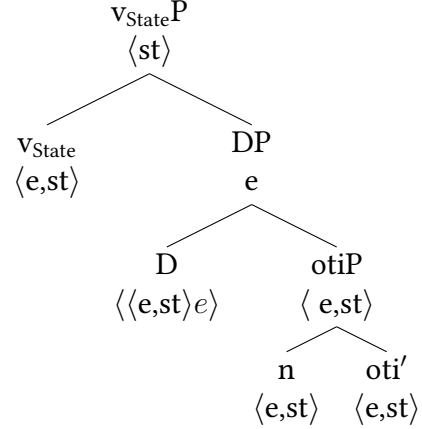
The proposed analysis of clausal complementation also has important implications for the mapping between Syntax and Semantics. To illustrate this, consider the observation that bare *oti*-clauses yield an *explanans* reading (the explanation itself), whereas nominalized *to oti*-clauses yield an *explanandum* reading (the fact being explained) (see 4). In Elliott (2020), these two readings can be derived via distinct compositional paths—semantic modification and argument saturation, respectively. Specifically, the *explanans* interpretation arises when the embedded clause is interpreted as a modifier of the verb through Predicate Modification (Heim and Kratzer 1998), while the *explanandum* interpretation emerges when the clause is interpreted as an argument of the verb via Functional Application (Heim and Kratzer 1998). Bondarenko (2022) further assumes a transparent Syntax-Semantics mapping, such that these two compositional paths are reflected in distinct syntactic configurations: bare embedded clauses are syntactic adjuncts (modifiers), whereas nominalized clauses function as arguments.

Although I do not pursue a full semantic analysis here, the present proposal is compatible with the view that the two readings (*explanans* vs. *explanandum*) reflect distinct compositional paths. Let us assume that clauses introduced by *oti*, which—as we have seen—express content, denote predicates of individuals with propositional content, of type $\langle e, st \rangle$. These clauses first combine semantically as modifiers—via Predicate Modification—with the light noun in their specifier, as illustrated in (56). This allows the *oti*P to retain its $\langle e, st \rangle$ semantic type. The resulting *oti*P then composes with the verb through the same modification relation, yielding the *explanans* reading, in line with Elliott (2020). The *explanandum* reading, by contrast, arises when a definite D merges on top of *oti*P, as in nominalized clauses, converting the *oti*-clause into an individual-denoting expression (see 57). This nominalized expression then combines with the matrix verb as its argument via Functional Application.

(56)



(57)



Our analysis, however, departs from Bondarenko (2022), who also assumes the same composition paths, on syntactic grounds. She argues that these distinct composition paths must be reflected in the syntax; therefore, clauses and DPs cannot occupy distinct syntactic positions relative to the verb—the first are merged as adjuncts whereas the latter as syntactic complements. However, as argued at length in Section 2, *oti*-clauses in Greek are true syntactic arguments (complements), even when they yield the *explanans* interpretation, and, crucially, they occupy exactly the same position as nominal arguments. If the compositional paths in (56) and (57) are correct, our findings suggest that the Syntax-Semantics mapping she assumes does not hold.

In other words, whether two elements combine via pure Merge (arguments) or Pair Merge (adjuncts) in syntax depends on their feature composition, which in turn determines their selectional properties—not their semantic denotations. Arguments are selected by their hosts, whereas adjuncts select their hosts. Since denotations are not accessible to Syntax, the system is blind to how elements will be semantically composed at LF.

This supports the autonomy of Syntax: the way syntactic structures are built in Syntax is independent of how they will be semantically composed at LF. Consequently, elements that are merged in the exact same syntactic position may nevertheless be semantically composed in different ways at LF.

8 Conclusion

This paper has argued for a new perspective on clausal complementation based on a detailed investigation of Greek *oti*- and *pu*-clauses. I have shown that a trio of puzzles—the internal-external argument asymmetry, the near-complementary distribution of the complementizers, and a novel stativity restriction—can be solved by a single, unified analysis. The core proposal is that C selects a light noun that must incorporate into the matrix verb. This mechanism, combined with a semantic distinction between content and situation and a bidirectional theory of selection for arguments and adjuncts, accounts for the full range of empirical facts. The analysis reinforces the view that selection is a fundamental licensing mechanism in Syntax and highlights the autonomy of the syntactic component from semantic interpretation.

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